Forces in Uniform Circular Motion

Answer key

Conventions used throughout

• **Newton's second law for circles:** $\Sigma F_c = ma_c = mv^2/r$. Also $a_c = 4\pi^2 r/T^2 = 4\pi^2 r f^2$.

• **Method (from the lesson):** draw the FBD, take +x in the direction of the acceleration (toward the centre), draw components, write a scalar equation for the net force (for example $F_{net} = T - F_g$), then use $\Sigma F_c = mv^2/r$.

• **Signs:** forces are written as sizes, with + for toward the centre (or up, for vertical balances) and − for the opposite way, as in the lesson's scalar equations.

• **Forces:** $F_g = mg$, F_N = normal force, T = tension, F_s = static friction (largest value $\mu_s F_N$).

• $g = 9.81\ m/s^2$. Speeds in km/h are changed to m/s (÷ 3.6). Every step keeps full precision; only final answers are rounded.

EXAMPLE 1

Car around a flat curve

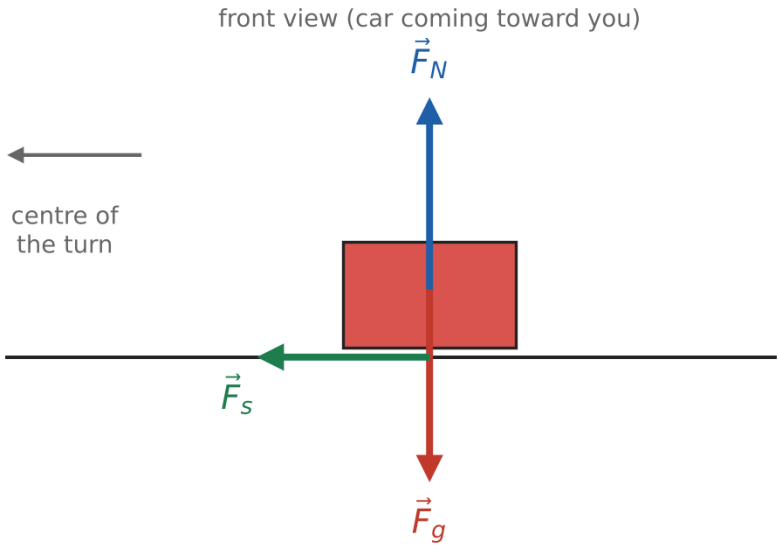
a) A car of mass m moving on a flat, horizontal road negotiates a curve. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.500, find the maximum speed the car can have and still make the turn successfully. DRAW THE FBD!

b) Based on the steps involved in this solution, why do highways not need multiple speed limits for the various masses of vehicles that travel along them?

(a) find the maximum speed the car can have and still make the turn successfully. DRAW THE FBD!

Static friction from the road is the only force toward the centre. At the maximum speed it is at its largest value, $\mu_s F_N$.

FBD of the car (front view)



GIVEN

$r = 35.0$ m $\qquad \mu_s = 0.500 \qquad v_{max} = ?$

Step 1 – Normal force (vertical balance)

$$\Sigma F_y = 0: \quad F_N - mg = 0 \quad \Rightarrow \quad F_N = mg$$

Step 2 – Friction supplies the centripetal force

Formula: $\Sigma F_c = \dfrac{mv^2}{r}$

Put in $\Sigma F_c = F_s = \mu_s F_N = \mu_s mg$: $\quad \mu_s mg = \dfrac{mv^2}{r}$

Divide both sides by m and multiply by r: $\quad v^2 = \mu_s gr$

Square root both sides:

$$v = \sqrt{\mu_s gr} = \sqrt{(0.500)(9.81)(35.0)} = 13.10 \text{ m/s}$$

(b) why do highways not need multiple speed limits for the various masses of vehicles?

The mass m cancelled in Step 2: $v_{max} = \sqrt{(\mu_s gr)}$. A heavier vehicle needs more centripetal force, but it also presses down harder and gets exactly that much more friction. The safe speed depends only on the road (μ_s and r), not on the vehicle's mass.

FINAL ANSWER

(a) v_{max} = 13.1 m/s

(b) The mass cancels, so the maximum speed is the same for every mass

Worksheet answer: 13 m/s
Why the answers differ: Same value, different rounding. √((0.500)(9.81)(35.0)) = 13.10 m/s. The data (35.0 m and 0.500) have 3 significant figures, so the key gives 13.1 m/s; the worksheet rounds to 2 figures.

EXAMPLE 2

Banking the curve

A civil engineer wishes to design a curved exit ramp for a highway in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a ramp is usually banked; this means the roadway is titled toward the inside of the curve.

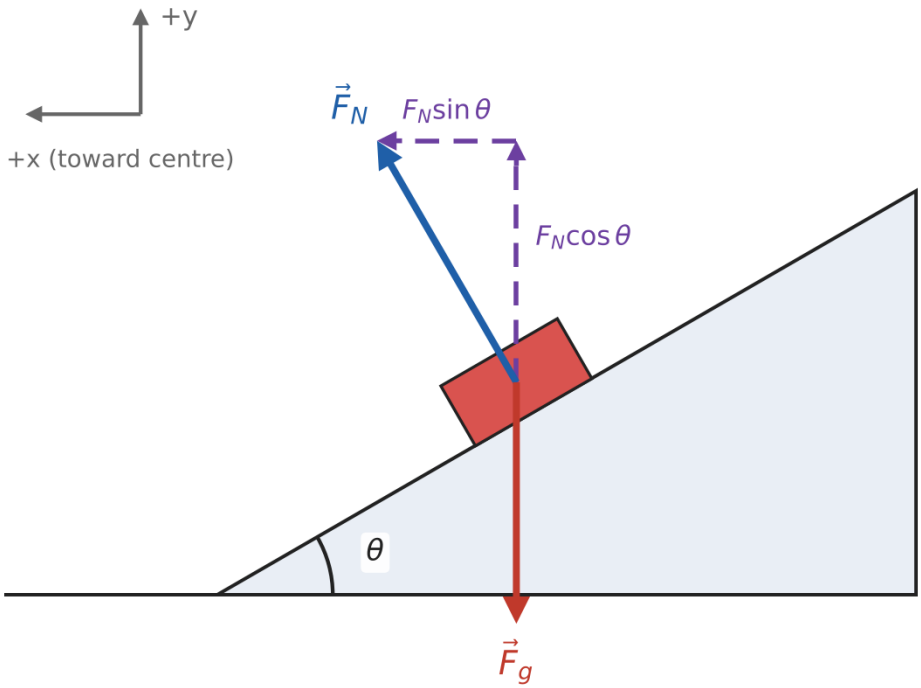
a) Suppose the designated speed for the ramp is to be 60 km/h and the radius of the curve is 50.0 m. At what angle θ should the curve be banked? (hint: think tan)

b) If a driver rounds the curve at a speed less than 60 km/h, which force (type and direction) would they need to rely on to keep them on the road? Draw an FBD showing this situation.

a) At what angle θ should the curve be banked? (hint: think tan)

No friction: only gravity and the normal force act. The normal force is perpendicular to the road, so it tilts toward the centre by θ. Split it into components on the lesson's axes (+x toward the centre, +y up):

FBD of the car on the banked ramp (no friction)



GIVEN

$v = 60 \text{ km/h} = 16.67 \text{ m/s}$ $r = 50.0 \text{ m}$

$\theta = ?$

Step 1 – Vertical balance

$$\Sigma F_y = 0: \quad F_N \cos\theta - mg = 0 \quad \Rightarrow \quad F_N \cos\theta = mg$$

Step 2 – Toward the centre

$$\Sigma F_x = \frac{mv^2}{r}: \quad F_N \sin\theta = \frac{mv^2}{r}$$

Step 3 – Divide the two equations (think tan)

Divide Step 2 by Step 1. F_N and m cancel, and sin θ ÷ cos θ = tan θ:

$$\frac{F_N \sin\theta}{F_N \cos\theta} = \frac{mv^2/r}{mg} \quad \Rightarrow \quad \tan\theta = \frac{v^2}{rg}$$

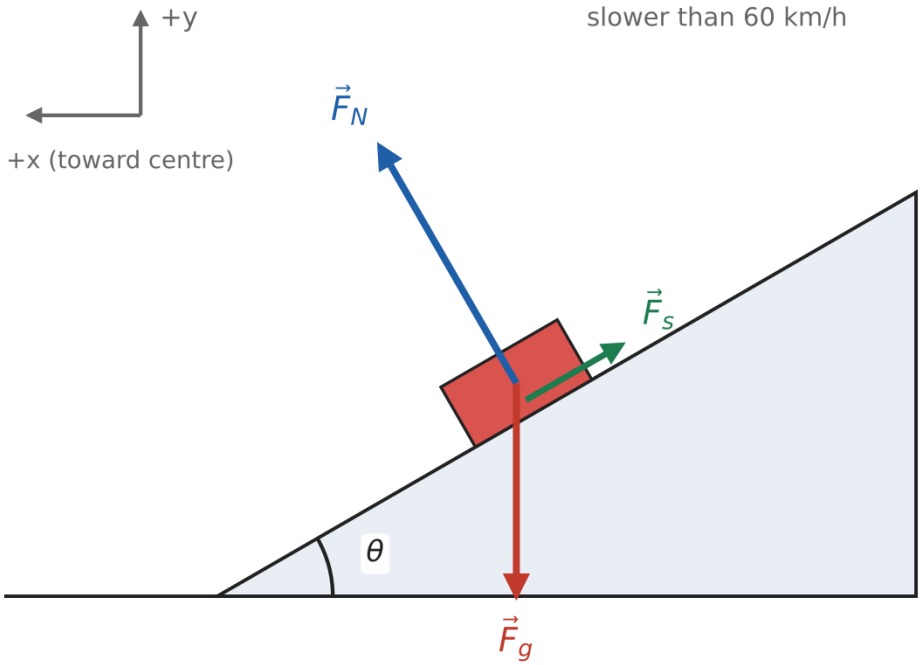
$$\tan\theta = \frac{(16.667)^2}{(50.0)(9.81)} = \frac{277.78}{490.5} = 0.5663$$

$$\theta = \tan^{-1}(0.5663) = 29.52°$$

b) If a driver rounds the curve at a speed less than 60 km/h, which force (type and direction) would they need to rely on to keep them on the road? Draw an FBD showing this situation.

Slower than the design speed, the car needs less centripetal force than $F_N \sin\theta$ gives, so it would slide down the bank toward the inside. Static friction points up the bank to stop that.

FBD of the car below the design speed



FINAL ANSWER

a) θ = 30°

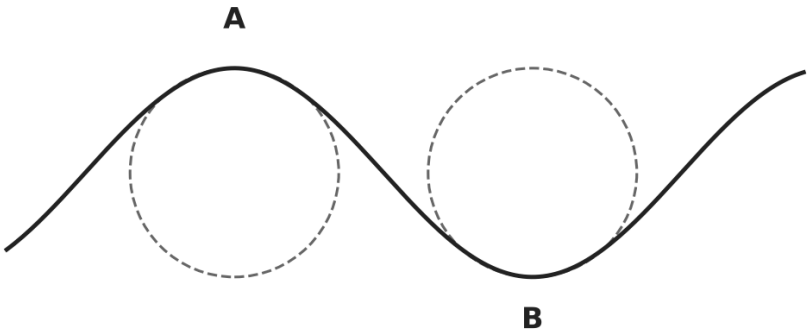
b) Static friction, pointing up the bank (away from the centre)

EXAMPLE 3

Roller coaster and apparent weight

1. Imagine you are on a roller coaster. Do you feel heavier than usual or lighter than usual...
   a. as you go over a hill (position A)? _____
   b. when you are at the bottom of a hill (position B)? _____
2. As you can see in the diagram, the top and bottom portions of hills can be approximated as circular motion. Draw a free body diagram of a person at position A and at position B.
3. What **direction** is the acceleration at A? _____ at B? _____
4. Based on your answer for 3, should the normal force be bigger or smaller than the force of gravity at A? What about at B? Fix your free body diagrams if necessary.
5. Say a person at the bottom of a hill on a roller coaster feels four times heavier than usual. Determine the speed of the roller coaster if the hill has a radius of curvature of 13.6 m.

Picture the situation



A is the top of a hill (the circle's centre is below the track); B is the bottom of a hill (the centre is above the track).

1. Do you feel heavier than usual or lighter than usual at A? At B?

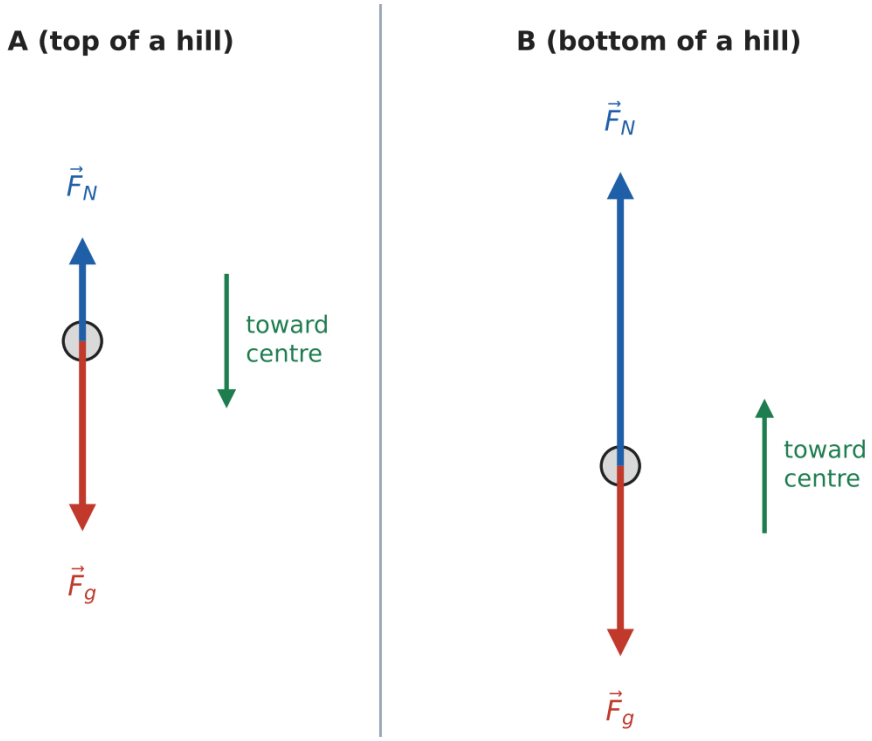
Going over the top of a hill you feel lighter (your stomach "lifts"). At the bottom you are pressed into the seat and feel heavier. Questions 3 and 4 show why.

2.–4. FBDs at A and B, the direction of the acceleration, and F_N compared with F_g

The acceleration points toward the centre of each circle: **down** at A and **up** at B. The net force points the same way, so at A gravity must be bigger than the normal force, and at B the normal force must be bigger than gravity:

$$\text{A:}\quad F_N = mg - \frac{mv^2}{r}\ (< mg) \qquad \text{B:}\quad F_N = mg + \frac{mv^2}{r}\ (> mg)$$

FBDs of the person (after checking the acceleration)



5. Determine the speed of the roller coaster if the hill has a radius of curvature of 13.6 m.

Feeling four times heavier means the seat pushes with $F_N = 4mg$. At B, + is up (toward the centre):

Formula: $\Sigma F_c = \dfrac{mv^2}{r}$

Put in the forces at B: $F_N - mg = \dfrac{mv^2}{r} \quad \Rightarrow \quad 4mg - mg = \dfrac{mv^2}{r}$

Divide both sides by m and multiply by r: $v^2 = 3gr$

$v = \sqrt{3(9.81)(13.6)} = \sqrt{400.25} = 20.006$ m/s

FINAL ANSWER

1. a. lighter; b. heavier

2. FBDs above

3. A: down (toward the centre); B: up (toward the centre)

4. A: $F_N < F_g$; B: $F_N > F_g$

5. v = 20.0 m/s

EXAMPLE 4

Swinging a mass in a horizontal circle

A 0.500 kg mass is swung on a 0.90 m long string in a horizontal circle so that the string makes an angle of 35° to the vertical. What is the speed of the mass around the circle?

a) Draw an FBD of the mass.

b) What would be the best direction for +x? Draw appropriate components.

VERTICAL FORCES:

c) Since the acceleration is horizontal, what happens to the vertical forces?

_____

d) Use your answer above to solve for the tension force.

HORIZONTAL FORCES:

e) What is the only horizontal force? _____

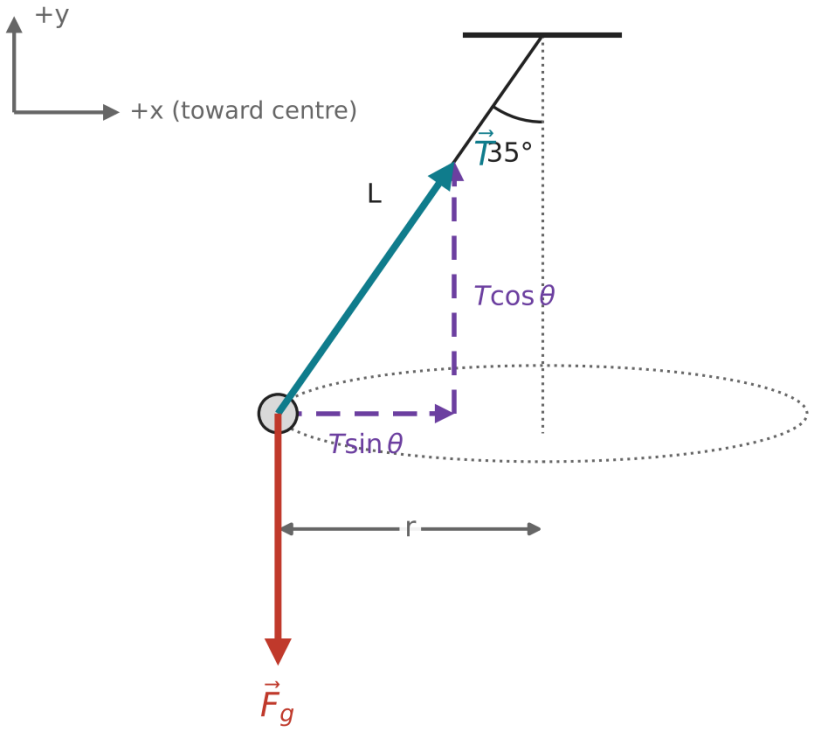
f) Use Newton's second law to solve for the centripetal acceleration.

g) Finally, find the speed from the centripetal acceleration. Remember, you are given the length of the string, but this is NOT the radius of the circle. You have to find r first. (See the diagram above.)

a), b) Draw an FBD of the mass. What would be the best direction for +x? Draw appropriate components.

Two forces: tension along the string and gravity. The acceleration points horizontally toward the centre of the circle, so +x is toward the centre and +y is up. Tension is the only force at an angle: T sin θ is horizontal (opposite the 35° angle at the top) and T cos θ is vertical (next to it).

FBD of the mass (+x toward the centre)



c) Since the acceleration is horizontal, what happens to the vertical forces?

There is no vertical acceleration, so the vertical forces balance (add to zero).

d) Use your answer above to solve for the tension force.

$\Sigma F_y = 0: \quad T\cos\theta - mg = 0$

Add mg and divide by $\cos\theta$: $\quad T = \dfrac{mg}{\cos\theta}$

$$T = \frac{(0.500)(9.81)}{\cos 35°} = 5.988 \text{ N}$$

e) What is the only horizontal force?

The horizontal component of the tension, T sin θ. It is the centripetal force.

f) Use Newton's second law to solve for the centripetal acceleration.

$\Sigma F_x = ma_c: \quad T\sin\theta = ma_c$

Divide both sides by m: $\quad a_c = \dfrac{T\sin\theta}{m}$

$$a_c = \frac{(5.988)\sin 35°}{0.500} = 6.869 \text{ m/s}^2$$

g) Finally, find the speed from the centripetal acceleration.

The radius is the horizontal distance from the centre to the mass, opposite the 35° angle:

$$r = L\sin\theta = (0.90)\sin 35° = 0.5162 \text{ m}$$

Formula: $a_c = \dfrac{v^2}{r} \quad \Rightarrow \quad v = \sqrt{a_c r} = \sqrt{(6.869)(0.5162)} = 1.883 \text{ m/s}$

FINAL ANSWER

a), b) FBD above, +x toward the centre

c) They balance ($\Sigma F_y = 0$)

d) T = 6.0 N

e) T sin θ, the horizontal part of the tension

f) $a_c = 6.9 \text{ m/s}^2$

g) v = 1.9 m/s

EXAMPLE 5

Swinging a mass in a vertical circle

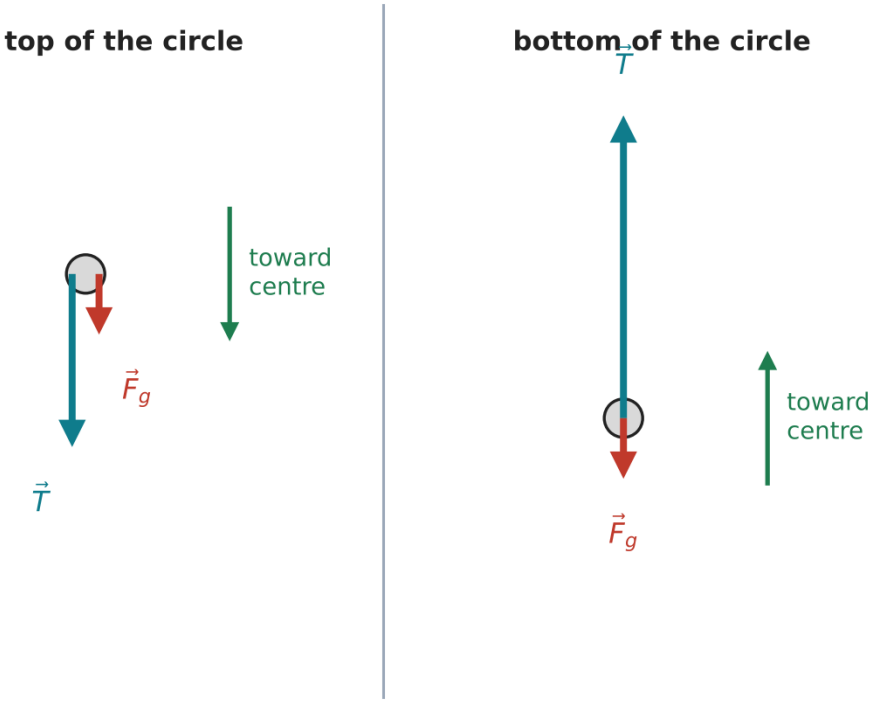
A 0.500 kg mass is swung on a 65 cm long string in a vertical circle with a frequency of 1.2 Hz.

a) Calculate the centripetal acceleration of the mass.

b) Find the tension in the string at the top of the circle. Draw an FBD.

c) The maximum tension the string can take is 20 N. Will the string break at any point during the swing? If so, will it happen near the top or the bottom?

FBDs of the mass (+ toward the centre)



a) Calculate the centripetal acceleration of the mass.

The frequency is given, so use the frequency form. r = 65 cm = 0.65 m:

$$a_c = 4\pi^2 r f^2 = 4\pi^2 (0.65)(1.2)^2 = 36.95 \text{ m/s}^2$$

b) Find the tension in the string at the top of the circle. Draw an FBD.

At the top, tension and gravity both point down, toward the centre:

$$\Sigma F_c = ma_c: \quad T + mg = ma_c$$

$$\text{Subtract } mg: \quad T = ma_c - mg$$

$$T = (0.500)(36.95) - (0.500)(9.81) = 18.476 - 4.905 = 13.571 \text{ N}$$

c) Will the string break at any point during the swing? If so, will it happen near the top or the bottom?

At the bottom, tension points up (toward the centre) and gravity points down, so the string must pull harder:

$$\Sigma F_c = ma_c: \quad T - mg = ma_c \quad \Rightarrow \quad T = ma_c + mg$$

$$T = 18.476 + 4.905 = 23.381 \text{ N}$$

That is more than the 20 N the string can take, so it breaks near the bottom.

FINAL ANSWER

a) a_c = 37 m/s^2

b) T = 14 N at the top

c) Yes, near the bottom, where T = 23 N > 20 N

Worksheet answer: b) 13.6 N
Why the answers differ: The physics is the same; only the rounding differs. The unrounded tension is 13.571 N, which is the worksheet's 13.6 N to 3 figures. The data (65 cm and 1.2 Hz) have only 2 significant figures, so the answer should be 14 N, like part (a), which the worksheet gives to 2 figures (37 m/s^2).

EXAMPLE 5

Minimum speed in a vertical circle

When you swing a mass on a string in a vertical circle, there is a minimum speed needed to maintain the circle. If you go too slow the mass just starts to FALL along a parabolic path inside the circular path it was following. Let's find that minimum speed.

In the problem you did above, you have only two forces. One force will not change no matter how slowly you swing the mass.

a) Which force will not change? _____

b) Which force will change? _____ What is the minimum possible value for this force? _____

A 0.500 kg mass is swung on a 65 cm long string in a vertical circle at the slowest speed needed to maintain circular motion. Draw the FBD for this situation.

a) What is the net centripetal force when it is at this lowest speed?

b) What is the lowest speed for the mass in this situation?

a), b) Which force will not change? Which force will change, and what is its minimum possible value?

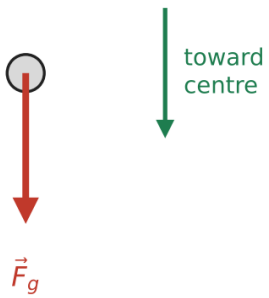
Gravity (mg) never changes. Tension changes with speed. A string can pull but never push, so the smallest tension is zero, which happens at the top at the slowest speed.

FBD Draw the FBD for this situation.

At the top at the slowest speed, T = 0, so gravity is the only force:

FBD at the top (slowest speed)

top, slowest speed



a) What is the net centripetal force when it is at this lowest speed?

$$\Sigma F_c = mg = (0.500)(9.81) = 4.905 \text{ N}$$

b) What is the lowest speed for the mass in this situation?

Formula: $\Sigma F_c = \frac{mv^2}{r} \quad \Rightarrow \quad mg = \frac{mv^2}{r}$

Divide by m, multiply by r, square root:

$$v = \sqrt{gr} = \sqrt{(9.81)(0.65)} = 2.525 \text{ m/s}$$

FINAL ANSWER

Gravity doesn't change; tension changes, and its minimum is 0 N

a) ΣF_c = 4.9 N (gravity alone)

b) v_{min} = 2.5 m/s

PRACTICE 1

Ball whirled in a horizontal circle

1. A 200. g ball on the end of a string is rotated in a horizontal circle of radius 10.0 m. The ball completes 10 rotations in 5.0 s. What is the centripetal force of the string on the ball? (316 N)

Frequency = rotations ÷ time. Then use the frequency form of a_c in $F_c = ma_c$:

$$f = \frac{10}{5.0\ \text{s}} = 2.0\ \text{Hz}$$

$$F_c = m(4\pi^2 r f^2) = (0.200)(4\pi^2)(10.0)(2.0)^2 = 315.8\ \text{N}$$

FINAL ANSWER

$F_c = 3.2 \times 10^2$ N toward the centre

Worksheet answer: 316 N
Why the answers differ: The physics is the same; only the rounding differs. The unrounded force is 315.8 N, which is the worksheet's 316 N to 3 figures. The time 5.0 s has only 2 significant figures, so the answer is 3.2×10^2 N.

PRACTICE 2

Electron in the Bohr model

2. In the Bohr model of the hydrogen atom, the electron revolves around the nucleus. If the radius of the orbit is 5.3 x 10^{-11} m and the electron makes 6.6 x 10^{15} rotations per second, find: a. the acceleration of the electron (9.1 x 10^{22} m/s^2) b. the centripetal force acting on the electron – due to the electrostatic attraction between the nucleus and electron. The mass of the electron is 9.1 x 10^{-31} kg. (8.3 x 10^{-8} N)

Rotations per second is the frequency:

a. $a_c = 4\pi^2 r f^2 = 4\pi^2(5.3 \times 10^{-11})(6.6 \times 10^{15})^2 = 9.114 \times 10^{22}$ m/s^2

b. $F_c = ma_c = (9.1 \times 10^{-31})(9.114 \times 10^{22}) = 8.294 \times 10^{-8}$ N

FINAL ANSWER

a. a_c = 9.1 × 10^{22} m/s^2 toward the nucleus

b. F_c = 8.3 × 10^{-8} N toward the nucleus

PRACTICE 3

Pendulum tension

3. A string pendulum 1.12 m long has a bob with a mass of 200 g. a. What is the tension in the string when the pendulum is at rest. (1.96 N) b. What is the tension at the bottom of the swing. If the pendulum is moving at 1.2 m/s^2? (2.2 N)

a. At rest, T balances gravity. b. "1.2 m/s^2" must mean a speed of 1.2 m/s (the answer 2.2 N uses that). At the bottom, T points up toward the centre and gravity points down:

a. $T = mg = (0.200)(9.81) = 1.962$ N

b. $T - mg = \dfrac{mv^2}{r} \quad \Rightarrow \quad T = mg + \dfrac{mv^2}{r}$

$$T = 1.962 + \frac{(0.200)(1.2)^2}{1.12} = 1.962 + 0.2571 = 2.219 \text{ N}$$

FINAL ANSWER

a. T = 1.96 N

b. T = 2.2 N

PRACTICE 4

Critical speed at the top

4. When you whirl a ball on a cord in a vertical circle, you find a critical speed at the top for which the tension in the cord is zero. This is because the force of gravity on the object itself supplies the necessary centripetal force. How slowly can you swing a 2.5 kg ball like this so that it will just follow a circle with a radius of 1.5 m? (3.8 m/s)

With T = 0, gravity is the whole centripetal force (as in Example 5). The mass cancels:

$$mg = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{gr} = \sqrt{(9.81)(1.5)} = 3.836 \text{ m/s}$$

FINAL ANSWER

v = 3.8 m/s

PRACTICE 5

Largest and smallest tension in a vertical circle

5. An object of mass 3.0 kg is whirled around in a vertical circle of radius 1.3 m with a constant velocity of 6.0 m/s. Calculate the maximum and minimum tension in the string. (1.1 x 10 ^{2} N, 54 N)

The needed centripetal force is the same everywhere. Gravity works against the string at the bottom (maximum tension) and helps it at the top (minimum tension):

$$\frac{mv^2}{r} = \frac{(3.0)(6.0)^2}{1.3} = 83.08 \text{ N} \qquad mg = (3.0)(9.81) = 29.43 \text{ N}$$

$$\text{Bottom:} \quad T - mg = \frac{mv^2}{r} \quad \Rightarrow \quad T = 83.08 + 29.43 = 112.51 \text{ N}$$

$$\text{Top:} \quad T + mg = \frac{mv^2}{r} \quad \Rightarrow \quad T = 83.08 - 29.43 = 53.65 \text{ N}$$

FINAL ANSWER

Maximum (bottom): T = 1.1 × 10^2 N

Minimum (top): T = 54 N

PRACTICE 6

Snoopy’s loop

6. Snoopy is flying his vintage war plane in a “loop the loop” path chasing the Red Baron. His instruments tell him the plane is level at the bottom of the loop and travelling with a speed of 180 km/h. He is sitting on a set of bathroom scales and notes that they read four times the normal force of gravity on him. What is the radius of the loop? (85 m)

180 km/h ÷ 3.6 = 50.0 m/s. The scale reads $F_N = 4mg$ at the bottom (as in Example 3, #5):

$$F_N - mg = \frac{mv^2}{r} \quad \Rightarrow \quad 3mg = \frac{mv^2}{r}$$

Divide by m and solve for r: $$r = \frac{v^2}{3g} = \frac{(50.0)^2}{3(9.81)} = 84.9 \text{ m}$$

FINAL ANSWER

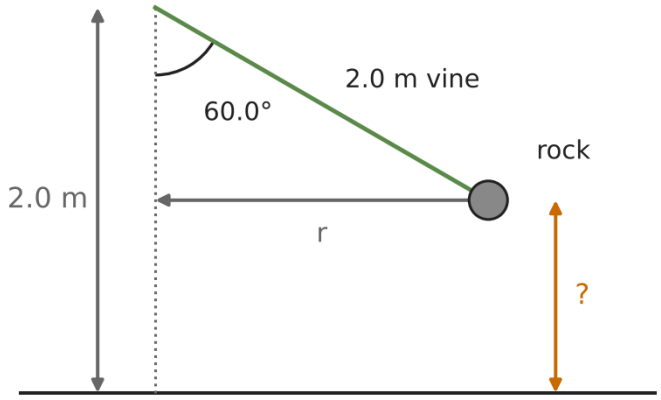
r = 85 m

PRACTICE 7

The kangaroo and the rock

7. An Australian bushman hunts kangaroos with the following weapon, a heavy rock tied to one end of a light vine of length 2.0 m. He holds the other end above his head, at a point 2.0 m above ground level and swings the rock in a horizontal circle. The cunning kangaroo has observed that the vine always breaks when the angle measured between the vine and the vertical reaches 60.0º. At what minimum distance from the hunter can the kangaroo stand with no danger of a direct hit? (2.4 m)

Side view at the moment the vine breaks



Step 1 – Radius and height of the circle

The radius is opposite the 60.0° angle (sine). The rock hangs below the hand by the part next to the angle (cosine):

$$r = (2.0)\sin 60.0° = 1.732 \text{ m} \qquad h = 2.0 - (2.0)\cos 60.0° = 1.0 \text{ m}$$

Step 2 – Speed when the vine breaks

This is the same set-up as Example 4: vertically T cos θ = mg, and toward the centre T sin θ = mv^2/r. Dividing gives $\tan\theta = v^2/(rg)$:

$$v = \sqrt{rg\tan\theta} = \sqrt{(1.732)(9.81)\tan 60.0°} = 5.425 \text{ m/s}$$

Step 3 – The rock as a projectile

When the vine breaks, the rock flies off horizontally along the tangent (first law) from 1.0 m up:

$$\text{Fall time:}\quad \Delta t = \sqrt{\frac{2\Delta d_y}{g}} = \sqrt{\frac{2(1.0)}{9.81}} = 0.4515 \text{ s}$$

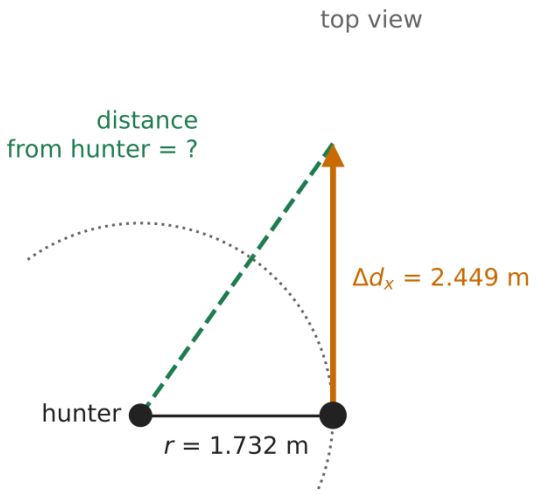
$$\text{Sideways distance:}\quad \Delta d_x = v\Delta t = (5.425)(0.4515) = 2.449 \text{ m}$$

Step 4 – Distance from the hunter

The rock starts 1.732 m from the hunter and flies off at 90° to that radius (along the tangent), so the two distances are at right angles:

$$D = \sqrt{r^2 + \Delta d_x^2} = \sqrt{(1.732)^2 + (2.449)^2} = 3.00 \text{ m}$$

Top view



FINAL ANSWER

The rock lands 3.0 m from the hunter, so the kangaroo is safe beyond about 3.0 m

(It travels 2.4 m from the point where the vine breaks.)

Worksheet answer: 2.4 m
Why the answers differ: 2.4 m is only the sideways distance the rock flies **after** the vine breaks (Δd_x = 2.449 m). The worksheet measures it as if the rock started at the hunter, but it starts on the circle, 1.732 m from him, and flies off along the tangent, at 90° to that radius. So it lands $\sqrt{(1.732^2 + 2.449^2)}$ = 3.00 m from the hunter. A kangaroo 2.4 m from the hunter is **not** safe: the rock passes that distance only about 0.5 m above the ground, low enough to hit it.

PRACTICE 8

Banking angle for a highway curve

8. A highway has a curve of radius 180 m. Traffic is expected to move around this curve at a speed of 100. km/h. What is the banking angle required for the road surface on the curve if there is no friction? (24º)

100. km/h ÷ 3.6 = 27.78 m/s. Use tan θ = $v^2/(rg)$ from Example 2:

$$\tan\theta = \frac{(27.778)^2}{(180)(9.81)} = 0.4370 \quad \Rightarrow \quad \theta = 23.60°$$

FINAL ANSWER

θ = 24°

PRACTICE 9

Speed that breaks the string

9. A 2.0 kg stone at the end of a string 0.50 m long is whirled in a horizontal circle. If the magnitude of the maximum tension that the string can withstand is 36 N, calculate the speed of rotation which would break the string. (3.0 m/s)

Treat the string as horizontal (for example, the stone sliding on a smooth table), so the whole tension is the centripetal force:

$$T = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{\frac{Tr}{m}} = \sqrt{\frac{(36)(0.50)}{2.0}} = 3.0 \text{ m/s}$$

If the stone hung freely in the air, the string would tilt (as in Example 4) and the speed would be about 2.5 m/s instead.

FINAL ANSWER

v = 3.0 m/s

PRACTICE 10

Girl on a swing

10. A 45 kg girl swings on a 2.4 m long light swing with two ropes. At the lowest point in her swing, the girl's speed is 5.0 m/s. Calculate the magnitude of the tension in each rope of the swing when the girl is at the lowest point of her swing. (460. N)

At the lowest point both ropes pull up (toward the centre) and gravity pulls down:

$$2T - mg = \frac{mv^2}{r} \quad \Rightarrow \quad 2T = mg + \frac{mv^2}{r}$$

$$2T = (45)(9.81) + \frac{(45)(5.0)^2}{2.4} = 441.45 + 468.75 = 910.2 \text{ N}$$

Divide by 2: $T = 455.1$ N

FINAL ANSWER

T = 4.6 × 10^2 N in each rope

Worksheet answer: 460. N
Why the answers differ: The tension is 455.1 N. The decimal point in "460." says the answer has 3 significant figures, and to 3 figures 455.1 N is 455 N, not 460 N. The data (45 kg, 2.4 m, 5.0 m/s) have only 2 significant figures, so the correct answer is 4.6 × 10^2 N. The worksheet's number is right to 2 figures, but its decimal point makes it wrong.

PRACTICE 11

Tension at the top of a vertical circle

11. A 0.50 m long rope is used to whirl a 1.0 kg mass in a vertical circle. When the mass was at the highest point, its speed was 2.5 m/s. Calculate the magnitude of the tension in the rope when the mass is at its highest point. (2.7 N)

At the top, tension and gravity both point down, toward the centre:

$$T + mg = \frac{mv^2}{r} \quad \Rightarrow \quad T = \frac{mv^2}{r} - mg$$

$$T = \frac{(1.0)(2.5)^2}{0.50} - (1.0)(9.81) = 12.5 - 9.81 = 2.69 \text{ N}$$

FINAL ANSWER

T = 2.7 N

PRACTICE 12

Car on an arched bridge

12. A bridge over a small river has a roadway which is in the shape of an arch having a radius of curvature of 41 m. What is the maximum speed at which an automobile can travel across the bridge without leaving the bridge? (20. m/s)

At the top of the arch the centre is below the car. Toward the centre: $mg - F_N = mv^2/r$. The car is about to leave the road when $F_N = 0$:

$$mg = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{gr} = \sqrt{(9.81)(41)} = 20.06 \text{ m/s}$$

FINAL ANSWER

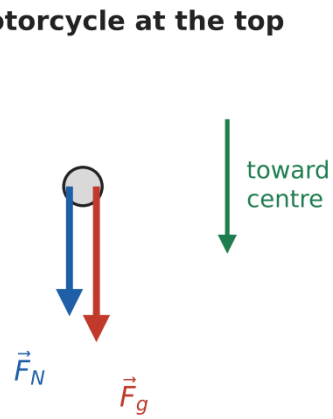
v_{max} = 20. m/s

PRACTICE 13

Motorcycle in a cylindrical track

13. A motorcycle and rider perform the stunt of circling in a cylindrical track with its axis horizontal. The diameter of the track is 10. m. The total mass of the rider and motorcycle is 200. kg. a. What is the force with which the motorcycle presses against the track at the highest point if the speed is 8.0 m/s? (6.0×10^2 N[up]) b. What is the minimum speed at the highest point if the motorcycle is not to fall from the track? (7.0 m/s)

FBD of the motorcycle at the top of the track



a.

r = 10. m ÷ 2 = 5.0 m. At the top, the track pushes the motorcycle down (toward the centre) and gravity also points down:

$$F_N + mg = \frac{mv^2}{r} \quad \Rightarrow \quad F_N = \frac{mv^2}{r} - mg$$

$$F_N = \frac{(200.)(8.0)^2}{5.0} - (200.)(9.81) = 2560 - 1962 = 598 \text{ N}$$

By Newton's third law, the motorcycle presses on the track with the same size of force, upward.

b.

The motorcycle just stays on the track when $F_N = 0$:

$$mg = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{gr} = \sqrt{(9.81)(5.0)} = 7.004 \text{ m/s}$$

FINAL ANSWER

a. 6.0×10^2 N [up] (motorcycle on the track)

b. v_{min} = 7.0 m/s

PRACTICE 14

Friction needed on a level curve

14. A car rounds a level curve of radius 65 m at a speed of 18 m/s. What is the minimum value of the coefficient of kinetic friction between the tires and the road to prevent skidding? (0.51)

As in Example 1, friction is the centripetal force: $\mu mg = mv^2/r$, so the mass cancels. (Tires that aren't skidding grip the road, so this is really the static coefficient.)

$$\mu = \frac{v^2}{gr} = \frac{(18)^2}{(9.81)(65)} = \frac{324}{637.65} = 0.5081$$

FINAL ANSWER

μ = 0.51 (minimum)

PRACTICE 15

Speed for a banked track with no friction

15. An auto track has semicircular ends of radius 150 m. The curves are banked at an angle of 52º. At what speed must the cars be driven to avoid any sidethrust on the wheels (i.e. so that no friction is required to maintain the turn)? (43 m/s)

Rearrange $\tan\theta = v^2/(rg)$ for v:

$$v = \sqrt{rg\tan\theta} = \sqrt{(150)(9.81)\tan 52°} = 43.40 \text{ m/s}$$

FINAL ANSWER

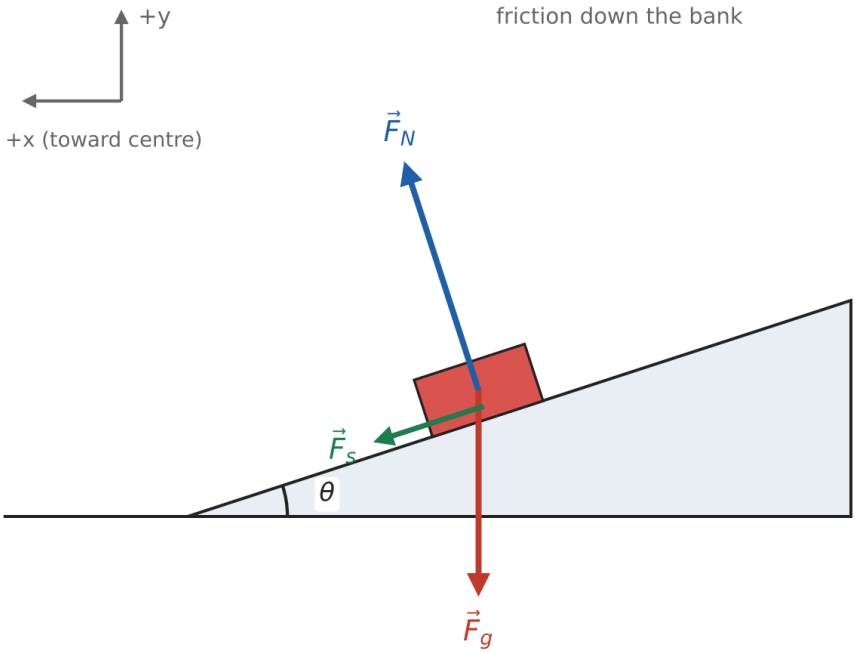
v = 43 m/s

PRACTICE 16

Banked curve with friction

16. A car having a mass of 1200 kg, rounds a curve which is banked at an angle of 18º to the horizontal. The radius of curvature is 130 m. Find the speed of the car if the frictional force between the tires and the road is 3.5 x 10^3 N down the banked curve. (29 m/s)

FBD of the car (+x toward the centre, +y up)



Step 1 – Components

F_N tilts toward the centre by θ: $F_N \sin\theta$ toward the centre, $F_N \cos\theta$ up. Friction points down the bank: $F_s \cos\theta$ toward the centre and $F_s \sin\theta$ down.

Step 2 – Vertical balance gives F_N

$$\Sigma F_y = 0: \quad F_N \cos\theta - F_s \sin\theta - mg = 0$$

$$\text{Solve for } F_N: \quad F_N = \frac{mg + F_s \sin\theta}{\cos\theta}$$

$$F_N = \frac{(1200)(9.81) + (3500)\sin 18°}{\cos 18°} = 13515 \text{ N}$$

Step 3 – Toward the centre gives v

$$\Sigma F_x = \frac{mv^2}{r}: \quad F_N \sin\theta + F_s \cos\theta = \frac{mv^2}{r}$$

$$13515 \sin 18° + 3500 \cos 18° = 4176 + 3329 = 7505 \text{ N}$$

$$v = \sqrt{\frac{(7505)(130)}{1200}} = 28.51 \text{ m/s}$$

FINAL ANSWER

v = 29 m/s

PRACTICE 17

The Rotor

17. Patrons on an amusement park ride called the Rotor stand with their backs against the wall of a rotating cylinder while the floor drops away beneath them. To keep from sliding downward, they require a centripetal acceleration in excess of about 25 m/s^2. The Rotor has a diameter of 5.0 m. What is the minimum frequency for its rotation? How does the person not fall out? (0.50 Hz)

r = 5.0 m ÷ 2 = 2.5 m. Solve $a_c = 4\pi^2 r f^2$ for f:

$$f = \sqrt{\frac{a_c}{4\pi^2 r}} = \sqrt{\frac{25}{4\pi^2(2.5)}} = 0.5033 \text{ Hz}$$

The wall pushes inward on each rider (normal force), and that push is the centripetal force. The faster the Rotor spins, the bigger the normal force, and so the bigger the static friction the wall can give. Static friction points up the wall and holds the rider up against gravity.

FINAL ANSWER

f = 0.50 Hz (minimum)

The wall's normal force gives the centripetal force; static friction up the wall balances gravity